

Audio, Image, and Video Data

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Overview

1 Audio

2 Image

3 Video

Sound as Data

Sound waves (as data) are a combination of two things:

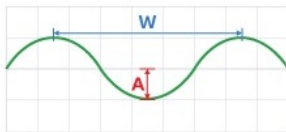
- Frequency: Pitch (low or high)
- Amplitude: Volume (low or high)



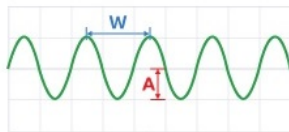
No sound



A loud sound with a low pitch



A quiet sound with a low pitch



A quiet sound with a high pitch



Amplitude



Wavelength

Audio Data Applications in Political Science

- What people say matters, but perhaps **how** they say it contains additional information
- **Audio recordings**: political debates, campaign advertisements, campaign speeches, legislative speeches, television news, talk radio, judicial deliberation.
- Challenges:
 - How to represent audio recordings numerically?
 - How to infer relevant quantities?
 - How to deal with temporal patterns (speech flow)?
- Method: **audio featurization**, requiring **preprocessing** into **utterances**
- See R packages `tuneR`, `seewave`, and `communication`

From Sound to Numbers: Audio Features

Key challenge: how do we represent the **texture** of speech — not just what was said, but **how** it was said?

- “**Audio as data**” **paradigm** leverages alignment between audio and transcripts (Dietrich, Enos, & Sen 2019; Dietrich, Hayes, & O’Brien 2019; Knox & Lucas 2021; Proksch, Wratil, & Wäckerle 2019; Rittmann 2024; Klostad 2016; but see Damann, Knox, & Lucas 2025)
 - Raw pitch, speaking rate, pause length, loudness
- More recently: low-level descriptors (**LLDs**), Mel-frequency cepstral coefficients (**MFCCs**), and modern **audio embeddings** (Mestre & Ryan 2026)
 - “Classic” techniques (**LLD**, **MFCC**): low-level, signal-based features like tone, emphasis, speaking style
 - “Modern” techniques (**embeddings**): complex, high-level / abstract attributes (e.g., emotional recognition); language-specific training;

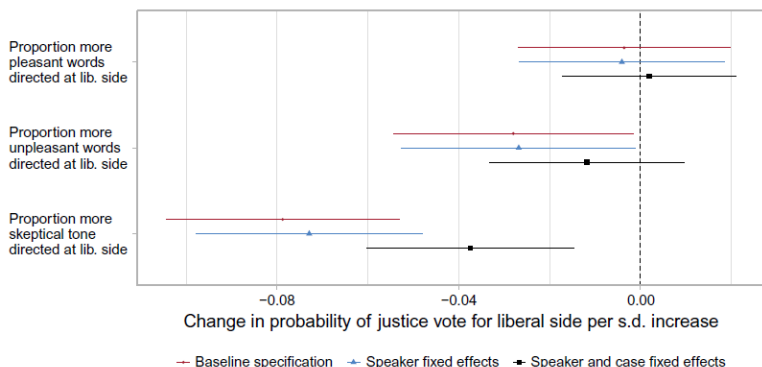
“Emotional Arousal” (Dietrich, Enos, & Sen 2019)

Table 1. Does vocal pitch predict votes in favor of the petitioner?

	Intercept only (1)	No controls (2)	DAL (3)	Harvard IV (4)	LIWC (5)
Fixed effects					
Constant	0.202 (0.056)	0.178 (0.055)	-0.025 (0.160)	-0.027 (0.160)	-0.026 (0.160)
Pitch difference		-0.266 (0.036)	-0.214 (0.038)	-0.215 (0.038)	-0.214 (0.038)
% More unpleasant words directed at petitioner			-1.971 (1.471)	0.071 (0.846)	-2.129 (1.308)
% More pleasant words directed at petitioner			-1.647 (1.086)	0.272 (0.685)	-1.673 (1.046)
# More questions directed at petitioner			-0.057 (0.008)	-0.057 (0.008)	-0.057 (0.008)
Political ideology _{t-1}			0.158 (0.033)	0.158 (0.033)	0.158 (0.033)
LC decision was conservative			0.011 (0.073)	0.012 (0.073)	0.012 (0.073)
Political ideology _{t-1} × LC decision was conservative			-0.263 (0.034)	-0.264 (0.034)	-0.263 (0.034)
Solicitor general as amicus supporting petitioner			0.540 (0.079)	0.543 (0.079)	0.540 (0.079)
Solicitor general as amicus supporting respondent			-0.672 (0.104)	-0.666 (0.104)	-0.673 (0.104)
# of amicus briefs supporting petitioner			0.039 (0.008)	0.039 (0.008)	0.039 (0.008)
# of amicus briefs supporting respondent			-0.058 (0.007)	-0.058 (0.007)	-0.058 (0.007)
Petitioner's level of resources			0.045 (0.014)	0.046 (0.014)	0.045 (0.014)
Respondent's level of resources			-0.003 (0.014)	-0.003 (0.014)	-0.003 (0.014)
Random effects					

“Model of Speech” (Knox & Lucas 2021)

FIGURE 4. Predicting Justice Votes with Directed Skepticism and Directed Affective Language



Note: Horizontal error bars represent point estimates and 95% confidence intervals from regressions of justice votes on directed pleasant words, directed unpleasant words, and our audio-based directed skepticism. Red circles correspond to a specification with no additional controls; blue triangles report results with speaker fixed effects only, black squares with speaker and case fixed effects.

Emotions in political debates (Mestre & Ryan 2021)

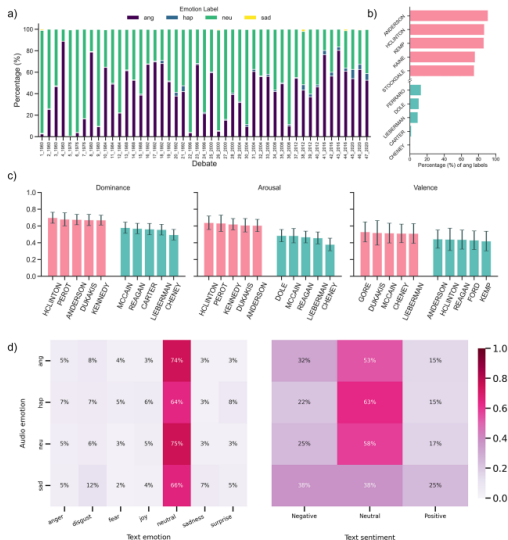


Figure 4. Emotional analysis using fine-tuned Wav2Vec 2.0 models on the U.S. presidential debates. (a) Percentage of discrete emotional labels (angry, happy, neutral, and sad) in each of the debates. (b) Top five and bottom five candidates ranked by their percentage of utterance classified as angry in their speeches. (c) Top five and bottom five candidates rated by their average dominance, arousal and valence. (d) Normalized co-occurrence matrices comparing audio-predicted emotions (rows) with text-based emotional classifications (left) and sentiment labels (right).

Overview

1 Audio

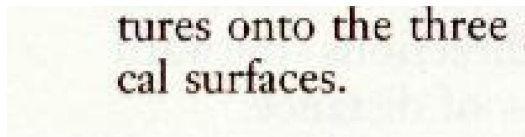
2 Image

3 Video

Images as Data

Images tend to be one of **two** types:

1. Raster images (pixels) – .bmp, .jpg, .jpeg, .png
2. Vector images (points, lines, and polygons) – .svg, .pdf, .eps



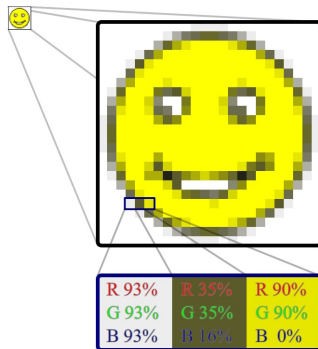
Source: <https://facul>

Note: Spatial data are actually a subset of image data!

Images as Data

We will focus on raster images, made up of three components:

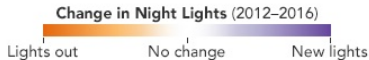
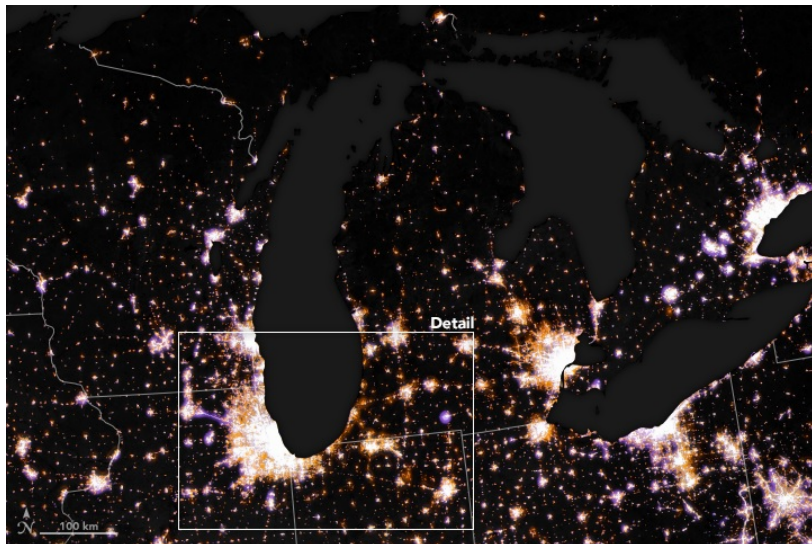
1. A grid of cells (equally sized)
2. A numerical value for each cell
3. A mapping of numerical values to colours (usually)



1. Remote sensing:

- **Electricity** (e.g., Baskaran, Min & Uppal 2014, Min et al 2013, Min et al 2014, Weidmann & Theunissen 2021)

Remote Sensing: NASA lights



Remote Sensing: NASA lights



Change in Night Lights (2012–2016)

Lights out

No change

New lights

1. Remote sensing:

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“Local Inequality” (Weidmann & Theunissen 2021)

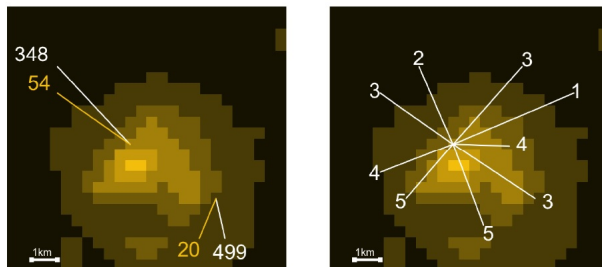


Figure 1. Satellite imagery of nighttime light emissions in raster format for the town Kansanshi in Zambia. **(Left panel):** The computation of the local Gini coefficient requires log-transformed nighttime light emissions at the level of cells (in yellow) and population estimates (in white). **(Right panel):** The DHS Wealth Index values of the households in the survey cluster at that location. All values are hypothetical and only displayed for illustration purposes.

Weidmann, N.B. and Theunissen, G. (2021). "Estimating Local Inequality from Nighttime Lights." *Remote Sensing*, 13(22), 4624.

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“Deforestation” (Baragwanath & Bayi 2020)

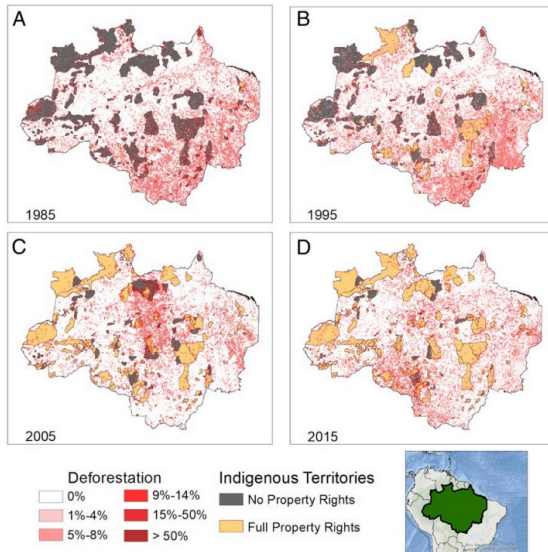


Fig. 1. Indigenous territories and deforestation in Brazilian Amazon for (A) 1985, (B) 1995, (C) 2005, and (D) 2015. Gray plots represent indigenous territories which have not received full property rights, while yellow plots represent plots which have received full property rights.

Image Data Applications in Political Science

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2. Image classification tasks:

- **Classifying media imagery** (e.g., Torres 2024)
- **Detecting anomalies in photographs** (e.g., Cantú 2019)
- **Identifying individuals in media images** (e.g., face detection in news photographs)

“Bag of Visual Words” (Torres 2024)

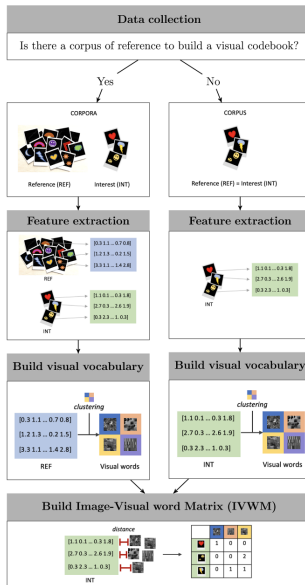


Figure 1. Workflow for building an Image-Visual Word Matrix.

“Bag of Visual Words” (Torres 2024)

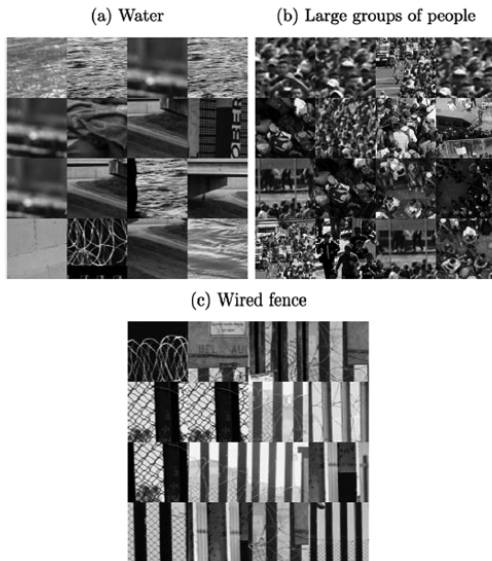


Figure 6. Examples of visual words.

“Fingerprints of Fraud” (Cantú 2019)

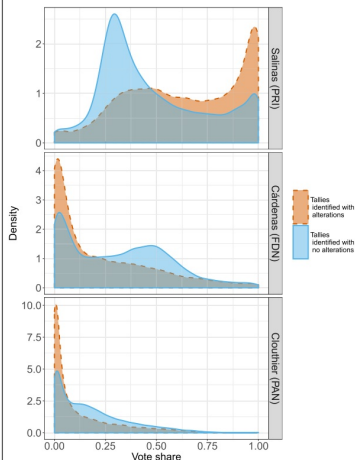
Francisco Cantú

FIGURE 1. Examples of Vote Tallies with Alteration in Their Numbers. Mexico, 1988



“Fingerprints of Fraud” (Cantú 2019)

FIGURE 4. Distribution of Vote Shares for Each of the Candidates. Mexico, 1988



Notes: The plots show the density distribution of the vote shares for the three main candidates of the 1988 election. Each line type corresponds to the classification of the vote tally sheet using the CNN classifier.

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- **Identifying features of physical / built environment that influence political participation** (e.g., Dasgupta & Rizzo Reyes 2026)
- **Identifying individuals in media images** (e.g., face detection in news photographs)

Built Environment (Rizzo & Dasgupta 2026)

FIGURE 1. Examples of Paired Street-level and Overhead Satellite Imagery of Polling Places



(a) Government Building in Aguascalientes



(b) Home in Mexico City



(c) School in Michoacan



(d) School in Sonora

Notes: Images represent pairs of Google Street View and Maps Satellite Imagery for four selected polling places from the dataset. Images were downloaded from the Google Maps API using latitude and longitude coordinates. All images are RGB and rescaled to dimensions 456×456 pixels.

Dasgupta, A. & Rizzo Reyes, T. (2026). Does the Built Environment Shape Voter Participation? Learning from Polling Place Imagery in Mexico. *American Political Science Review*.

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3. Optical character recognition (OCR)

Optical Character Recognition (OCR)

- Many political **documents** exist as images: scanned newspapers, historical records, ballot papers, legislative transcripts
- **OCR** converts images of printed or handwritten text into machine-readable text
- **Pipeline:** **image** → preprocessing → character recognition → **text**
 - The output can then be analyzed with standard text-as-data methods, but note that **accuracy may vary with document quality**
- **Applications** in political science:
 - Digitizing historical newspapers and campaign materials
 - Extracting text from administrative and legislative records
 - Processing scanned party manifestos and policy documents

Overview

1 Audio

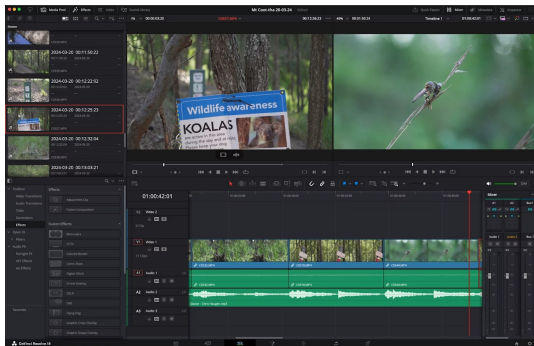
2 Image

3 Video

Video vs. Image/Audio Data

Video data represents a **combination of image** data and **audio** data:

- Sequence of still images
- Sequence of sounds
- Played back at some known speed, in unison



Video Data Applications in Political Science

1. Physical behavior from video footage:

- Racial avoidance in urban spaces (e.g., Dietrich & Sands 2023)
- Social distancing compliance from CCTV (e.g., Bernasco et al. 2023)
- Physical proximity in social interaction from 3D cameras (e.g., Goldstein, Legewie & Shiffer-Sebba 2023)

2. Nonverbal communication and facial expression:

- Candidate emotional expression in election debates (e.g., Boussalis, Coan, Holman & Müller 2021)
- Speech emphasis and gesture in legislative video (e.g., Rittmann, Ringwald & Nyhuis 2025)
- Emotional expression of populist leaders on YouTube (e.g., Major & Tomašević 2025)

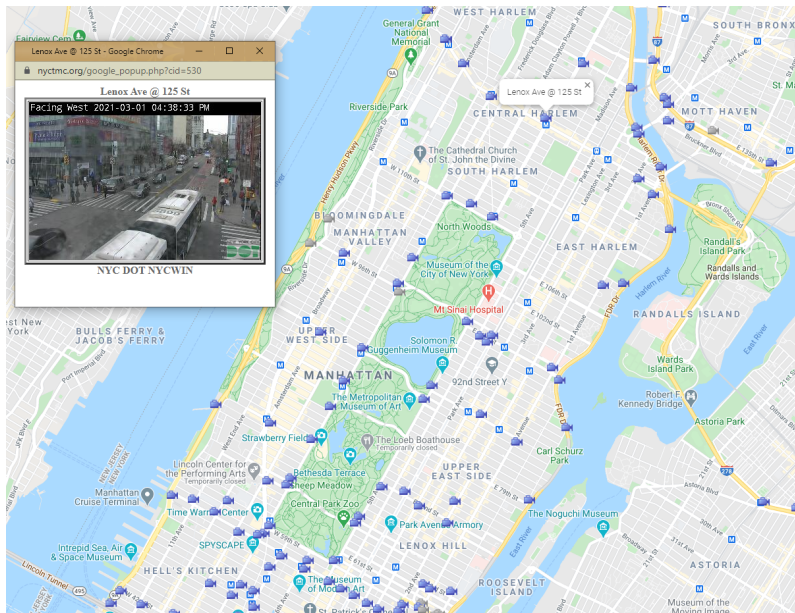
3. Broadcast and archival video:

- Automated coding of campaign advertisements (e.g., Tarr, Hwang & Imai 2023)
- Social polarization via motion detection in C-SPAN (e.g., Dietrich 2021)

4. Public deliberation:

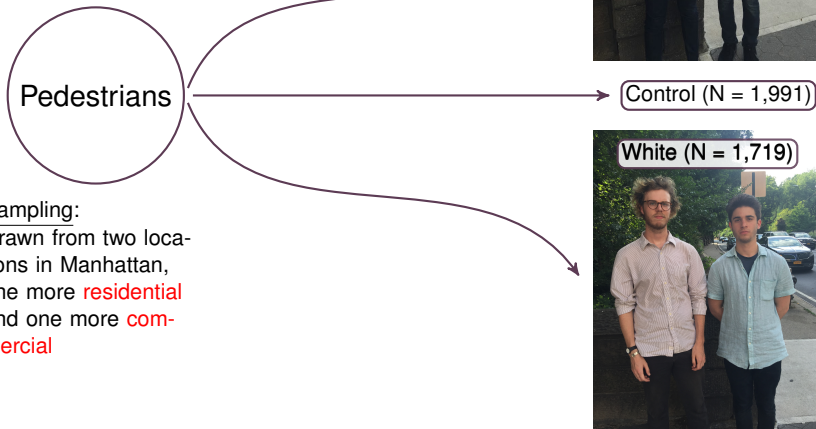
- Local government meeting recordings (e.g., Barari & Simko 2023)

“Racial Avoidance” (Dietrich & Sands 2023)



Randomization:

- Divide each day into **45-minute blocks**
- Divide each block into **15-minute clusters**
- **Block-randomize** clusters to conditions



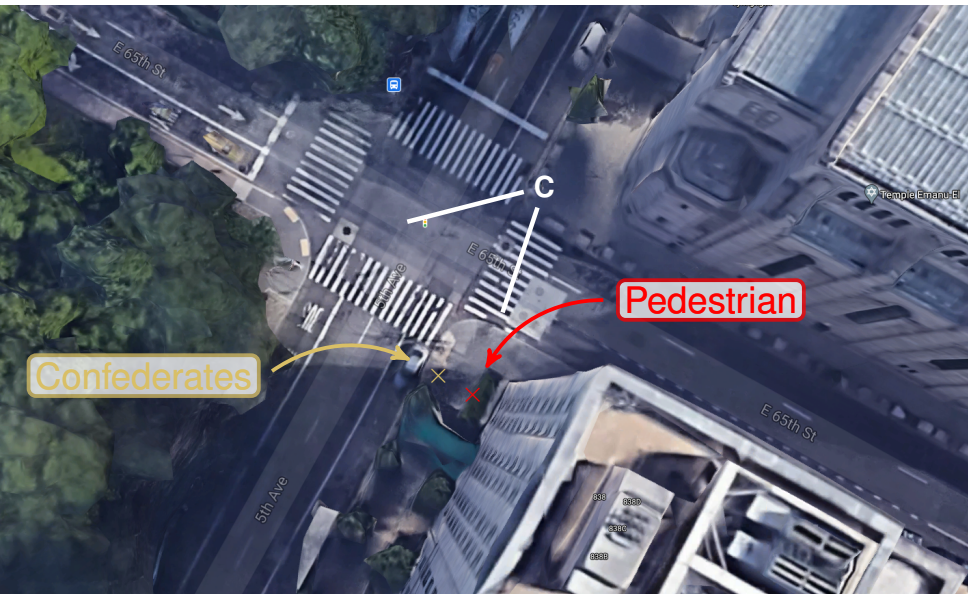
Sampling:

Drawn from two locations in Manhattan, one more **residential** and one more **commercial**

Facing East

2017-07-10 13:46:19





Facing East

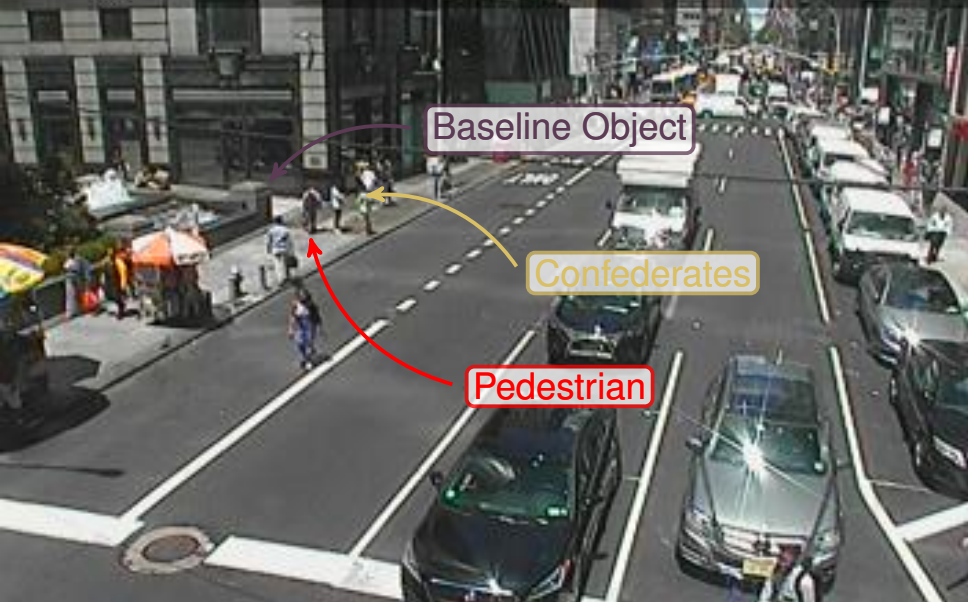
2017-07-10 13:46:19

Baseline Object

Pedestrian

Confederates

Facing East 06/20/2017 01:46:21 PM



From Video to Data

RAs record **pixel coordinates** for **each pedestrian**, **confederates**, and **baseline objects**.

imageJ

Tracking

experiment	pedestrian_id	track	slice	x	y	block	condition	wall_distance	confederate_distance	distance_diff	inside_box
lexington	lexington_1	1	1	83	99	1	1	21.023796	38.013156	16.9893596	0

Facing East

2017-07-10 16:29:04



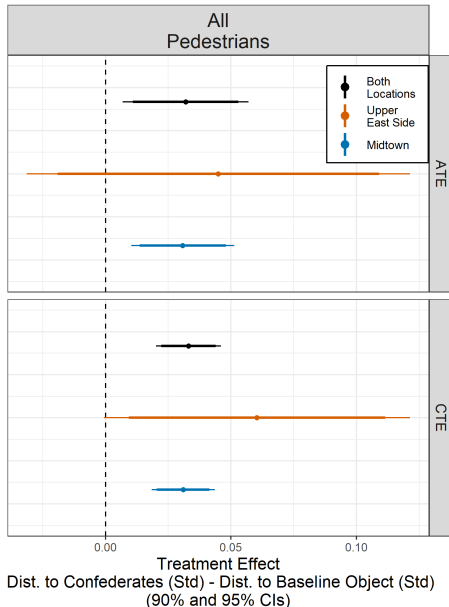
Facing East

06/20/2017 01:11:26 PM

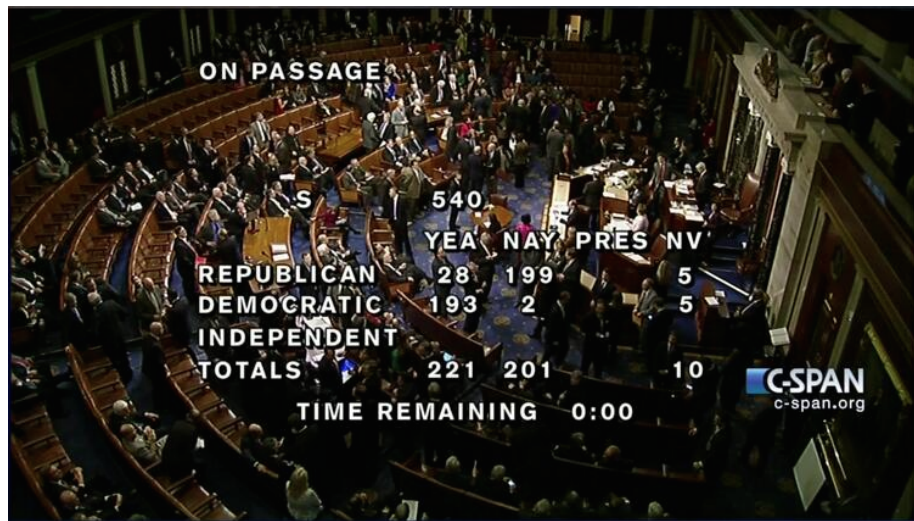


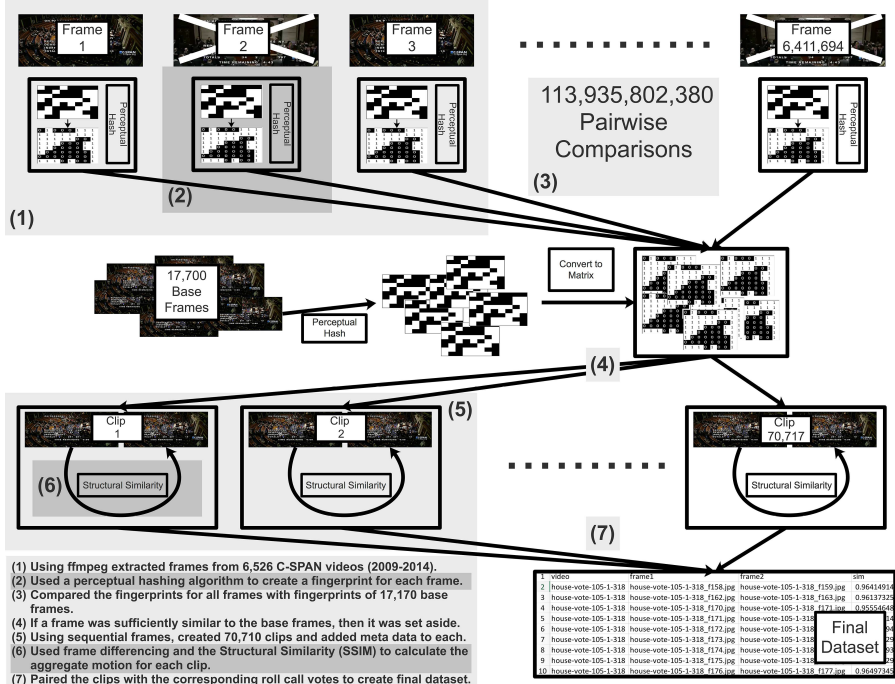
lexington	lexington_19	19	5	111	89	1	1	13.038405	14.866069	1.8276639	1
lexington	lexington_19	19	6	106	92	1	1	8.246211	17.000000	8.7537887	1

“Racial Avoidance” (Dietrich & Sands 2023)

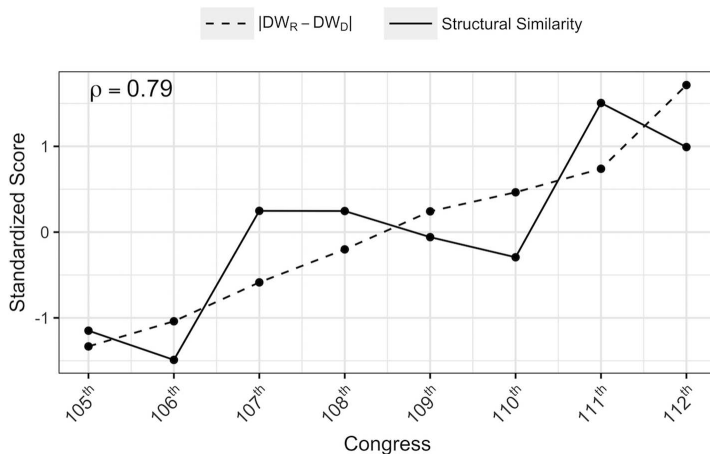


"Crossing the Aisle" (Dietrich 2021)





Polarization & SSIM Over Time (Dietrich 2020)



Dietrich, Bryce J. "Using motion detection to measure social polarization in the US House of Representatives." *Political Analysis* 29, no. 2 (2021): 250-259.

“LocalView” (Barari & Simko 2023)

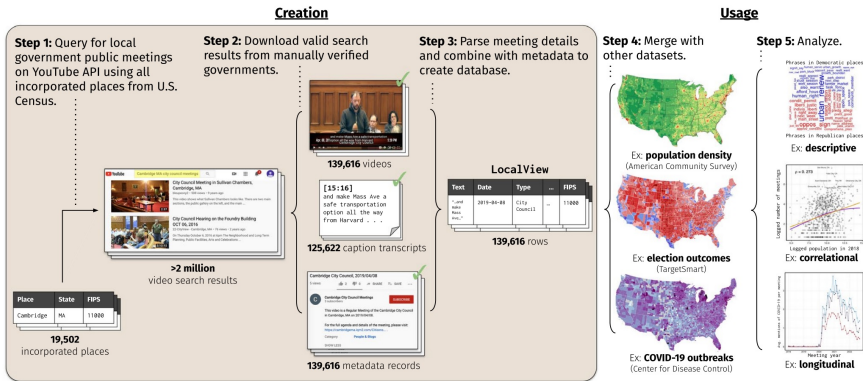


Fig. 2 Graphical Overview of Database Creation and Usage. Creation steps are performed on a regular cadence by the maintainers of the database; usage steps demonstrate how users may conduct analyses on the data, once it is downloaded.

Barari, S., & Simko, T. (2023). LocalView, a database of public meetings for the study of local politics and policy-making in the United States. *Scientific Data*, 10(1), 135.